

Universal Data Acquisition and Control with Yuktix UDAC for the IoT



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The Internet of Things (IoT) applications capture data from the physical world, communicate this data to a central server and run analytics on it. In order to complete the IoT loop, the user has to select sensors, choose a compatible platform to plug in those sensors, write the firmware, select data protocols, establish communication to the server and ensure security of the transferred data.

Problem arises when the user wants data related to machine utilisation, power consumption, temperature of the system and excess vibration in machines, and store this data on a computer system, access it on a mobile phone, get instant alerts and share these reports with other concerned people.

The machine-to-machine (M2M) market today is very fragmented. Creating such systems requires disparate skills from multiple vendors. It is a time-consuming process and the budget cannot be calculated beforehand. So how can we make the system-creation process simple, know what will one receive and at what cost?

The solution

The Yuktix Universal Data Acquisition and Control (Yuktix UDAC) platform comprises hardware and software that connect industrial equipment, machines and sensors to

the Internet in a plug-and-play model and let users control field devices from a central dashboard. It can be used for a variety of purposes including the following:

1. Connect industrial protocol equipment to the Internet, customer

relationship management and enterprise resource planning systems

2. Collect physical data such as temperature, vibration, stress and pressure
3. Establish solar-powered off-grid units to collect sensor data
4. Control units deployed in the field from a central location
5. Take decisions based on sensor data

Given below is a step-by-step guide to how the solution works:

1. Select the sensor(s) required for an application from the catalogue of compatible sensors provided
2. Plug the selected sensor(s) on the small form factor board
3. Upload the firmware provided
4. Power the board using a solar panel, lithium-polymer battery or mains electricity
5. Connect it to the cloud gateway using either a wired (Ethernet, RS485, etc) or a wireless (GSM, ZigBee, etc) solution
6. Start getting data collected by the sensors on Yuktix cloud, Yuktix Android app or external cloud services such as Amazon Web Services (AWS) and Zoho

More features

Yuktix has a secure digital (SD) card memory onboard. There is an external random access memory (RAM) that increases the power of onboard processing. It also has general-purpose input/output pins to interface with multiple digital and analogue devices and sensors. By powering the device on battery or solar energy and transmitting data over Global System for Mobile Communications (GSM), you can make a true off-the-grid system.

You can also set up a wireless (ZigBee) sensor mesh network with solar power in order to collect physical data from regions that have no network coverage or even electricity.

Buying a highly-efficient development board with varied functions would not be of

Yuktix board

